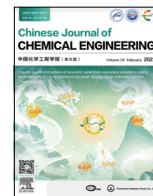




Contents lists available at ScienceDirect

Chinese Journal of Chemical Engineering

journal homepage: www.elsevier.com/locate/CJChE

Full Length Article

Eco-friendly biodegradable polyurethane based coating for antibacterial and antifouling performance

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ARTICLE INFO

Article history:

Received 31 December 2021

Received in revised form 20 September 2022

Accepted 22 September 2022

Available online 29 September 2022

Keywords:

Caprolactone

PEG segments

Biodegradable polyurethane

Biofouling

Hydrolytic degradation

ABSTRACT

Biofouling, which comprises the absorption of proteins and the adhesion of bacteria to the surface of living entities, is a severe concern for the maritime sector since it ultimately leads to hydrodynamic drag, resulting in a higher increase in fuel consumption. As a result, polymer resins are crucial in the marine sector for anti-biofouling coatings. In this work, the poly(caprolactone-ethylene glycol-caprolactone)-polyurethane (PECL-PU) are prepared through ϵ -caprolactone (CL), poly(ethylene glycol) (PEG), 4,4'-methylene bis(cyclohexyl isocyanate) and 1,4 butanediol. Our study demonstrate that the PECL-PU copolymer degraded in artificial seawater (5.21%), enzymatic solution (12.63%), and seawater (13.75%) due to the presence of PEG segments in the laboratory-based test under static condition. Because the addition of PEG segments are increased the polymer's amorphous area and decreased the crystallization of the polycaprolactone (PCL) in the copolymer, as demonstrated by differential scanning calorimetry, X-ray diffraction, and water contact angle studies. Therefore, the hydrolysis rates of PECL-PU were higher than the caprolactone-co-polyurethane (CL-PU). The antifouling test showed that PECL-PU3 copolymer had about 90.29% protein resistance, 85.2% *Escherichia coli* (*E. coli*) reduction and 94.61% marine diatom *Navicula incerta* reduction comparison to the control. We have developed an eco-friendly and inexpensive promising degradable polyurethane for reduction of bacterial biofilm, which can preserve the formation of biofouling on marine coating under practical sea conditions.

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1. Introduction

The submergence of a surface into an aquatic or freshwater atmosphere can contribute to adhesion within min of biomolecules including proteins, polysaccharides, and proteoglycans. The substance adhered serves as a conditioner for diatoms and marine bacteria that bind to it in the other fouling stage and emerge within 24 h after submersion. The attached bacteria would subsequently form a microbiological biofilm that facilitates the colonization of microalgae spores, promoting the accumulation of macroorganisms [1–6]. The attachment and resulting settlement on the surface produce a variety of issues, such as increasing fuel consumption owing to corrosion intensity and enhancing drag fraction due to

abrasion of the layer [7]. The change in surface roughness is triggered primarily via macro-organism and can be extremely high costs in both money and time [8].

Meanwhile, several strategies have been implemented to combat fouling in the marine environment. The traditional approach to mitigation biofouling to maritime vessel activities has been using antifouling coatings and paints [9]. Biocidal antifouling paints are the most common strategy, which works by releasing biocides or another toxin via leaching or ablative process in the immediate surroundings of a ship's hull. The usage of tributyltin-based self-polishing polymer (TBT-SPC) is responsible for rises in organotin rates around shipping lanes, harbors, and dry docks due to their release rate [10]. As a result of the poisoning and depletion of the fish population, the International Maritime Organization (IMO) prohibited the use of TBT-based paint on a worldwide scale [11].

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However, TBT-based approaches encourage the creation of environmentally friendly antifouling solutions, including liquid-infused slippery surfaces [12,13], silicone-based fouling release coatings (FRC) [14,15], amphiphilic copolymers [16–18], biodegradable polymers with complex surfaces [19,20].

However, the self-polishing acrylate copolymer and fouling release (FR) were also used after TBT based coating. The antifouling performance of self-polishing acrylate copolymer in the side chain with the hydrolyzable ester unit and FR depend on seawater flow and sailing speed of a ship there are not appropriate for slow ships and for the ship spending long time in port. In the sluggish ships or those in harbors, these coatings generally show weak antifouling coating for long time under static condition [21,22]. Among them, biodegradable polymers have been identified as antifouling materials in recent years owing to their sustained release rate and the enzymatic effect in the marine ecosystem under static condition [23]. Biodegradable polymers decompose into smaller monomeric or oligomer elements due to the influence of cleavable chemical bonds in the polymer backbone [24]. However, conventional polyesters such as polybutylene succinate (PBS), polylactic acid (PLA), and polycaprolactone (PCL) have a greater crystalline nature, poor adhesive to surfaces and a comparatively sluggish rate of release [25,26].

On the other hand, it is well known that poly(ethylene glycol) (PEG) or polysaccharides, poly(2-oxazoline), and zwitterions may impede the adsorption of proteins and microorganisms in aqueous solutions [26–32]. Mechanical properties of protein-resistant polymers are often poor since their hydrophilic tendency results in matrix swelling. Additionally, the dynamic maritime environment degrades the antifouling effectiveness of protein-resistant surfaces over time. Therefore, the surfaces cannot be replaced and are easily covered by some inorganic substances, leading to decrease their anti-biofouling efficiency [33–35]. The regeneration of protein-resistant polymer surfaces after damage or contamination, especially in complex marine environments, is a significant problem.

Several researchers in the biomedical and marine field have reported biodegradable polyurethanes based on PCL copolymers [36,37]. They have also described the formation of biodegradable polyurethanes based on PCL and PEG diblock or triblock copolymers and have investigated the relationships between the rate of biodegradation and the chemical structure of such polyurethanes [38–41]. They indicated that PCL improves the elastomeric mechanical performance and crystallinity of polyurethanes, whereas PEG might enhance aqueous absorption and PEG hydrophilicity. As a result, the reduction in PCL content and the increase in PEG content in the polymer contribute to higher rates of degradation [42]. Similarly, protein resistant polymer (PEG_{600/1000}) is also used as a carrier of the drug in the biomedical field [43]. Chemically incorporating PEG into a biodegradable chain is intended to prolong release while also improving antifouling ability and service life. Therefore, the combination of PCL with PEG_{600/1000} is hoping to show long-term antifouling and eco-friendly performance due to degradation rate (peel-off scales) in a complex marine environment [26,44].

In this work, we synthesized a biodegradable polyurethane (poly(caprolactone-ethylene glycol_{600/1000}), PECL) by ring open polymerization and condensation. We firstly synthesized the amphiphilic PECL with PEG_{600/1000} segments and PCL. The PEG segments demonstrate good hydrolytic degradation with dual antifouling activity. The biodegradable polyurethanes were then synthesized using key components, the complex PECL, 1,4 butanediol (1,4 BD), and 4,4'-methylene bis(cyclohexyl isocyanate) (H₁₂MDI). These antifouling polymers can be used as an effective alternative to toxic material. The adhesion and wettability test, hydrolytic degradation, and antifouling performance of biodegradable polyurethane surfaces have been studied.

2. Experimental

2.1. Materials

All the chemical were used without purification. ϵ -caprolactone (CL, 99%), 1,4 butanediol (1,4 BD, 99%), poly(ethylene glycol) MW 600/1000 (PEG_{600/1000}), *n*-hexane (97%), tetrahydrofuran (THF, 99%), toluene anhydrous (99.5%), dibutyltin dilaurate (DBTDL, 97%), dichloromethane (CH₂Cl₂, 99.5%), absolute methanol (99.9%), and lipase enzyme were bought from Aladdin, China. 4,4'-Methylene bis(cyclohexyl isocyanate) (H₁₂MDI, 90%) was bought from Merck, China. Stannous octoate (Sn(Oct)₂, 95%) was bought from Macklin, China. The artificial seawater (ASW, pH = 8.2) was prepared by sodium chloride (NaCl), sodium sulfate (Na₂SO₄), magnesium chloride hexahydrate (MgCl₂·6H₂O), potassium bromide (KBr), sodium fluoride (NaF), calcium chloride (CaCl₂), sodium bicarbonate (NaHCO₃), potassium chloride (KCl), boric acid (H₃BO₃), and strontium chloride hexahydrate (SrCl₂·6H₂O), while pH adjusted by 1 mol·L⁻¹ NaOH and HCl [45]. Seawater was obtained from Lincheng, Zhoushan, China.

2.2. Synthesis of poly(caprolactone-co-ethylene glycol_{600/1000}) (PECL)

The PECL prepolymer was synthesized by the modification of Lida et al. [43] method. To begin, 5.1363 g (45 mmol) CL was dissolved in 25 ml anhydrous toluene in a 200 ml three-necked flask and agitated at 150 r·min⁻¹ using a magnetic stirring plate. The CL suspension was mixed with 1.6 g (1.0 mmol) of PEG_{600/1000} and the mixture was magnetically agitated for 10 min at a speed of 200 r·min⁻¹. Separately, 0.0067 g (0.7% respect of total reactant) of Sn(Oct)₂ was dissolved in 15 ml anhydrous toluene in a schlenk tube and added dropwise to the CL/PEG_{600/1000} mixture. The mixture was deoxygenated by nitrogen flow for 15 min at room temperature (25 °C) and then heated at 180 °C at 250 r·min⁻¹. After 3.5 h, the PECL prepolymer was obtained. The raw product was precipitated in absolute methanol. In order to completely eliminate the soluble trace monomer from the precipitate, it was washed three times with CH₂Cl₂. The remaining prepolymer was dried under vacuum at 40 °C for 24 h.

2.3. Synthesis of biodegradable polyurethane (PECL-PU)

A two-step procedure was used to synthesize the biodegradable polyurethane using PECL (Fig. 1). In the first stage, 3.672 g (14 mmol) of H₁₂MDI was dissolved in 15 ml THF in a 500 ml three-necked flask and stirred at 250 r·min⁻¹ for 10 min. PECL (as-synthesized mentioned above) was dissolved in 12 ml THF by magnetic stirring for 5 min at 280 r·min⁻¹. Next, The PECL solution was then mixed with the H₁₂MDI suspension. The mixture was deoxygenated by nitrogen flow for 15 min at room temperature and then heated to 80 °C at 300 r·min⁻¹. After 1 h, the flask cooled at room temperature. In the 2nd step, 1.261 g (14 mmol) of 1,4 BD was dissolved in 10 ml THF and added to the PECL/H₁₂MDI solution. Separately, 0.1166 g (1% (mass) respect of total reactant) DBTDL (catalyst and extender) was dissolved in 15 ml THF in a schlenk flask and added drop by drop to the PECL/H₁₂MDI/1,4 BD mixture under room temperature and nitrogen flow and then heated at 85 °C at 300 r·min⁻¹. After 3 h, the PECL-PU polymer was obtained. The polymer was entirely precipitated in the excess amount of *n*-hexane, then filtered and dried under vacuum at 40 °C for 24 h. This was repeated three times to purify the polymer. Four biodegradable polyurethane (CL-PU, PECL-PU1, PECL-PU2, and PECL-PU3) were synthesized with different molar ratios, which was showed in Table 1.

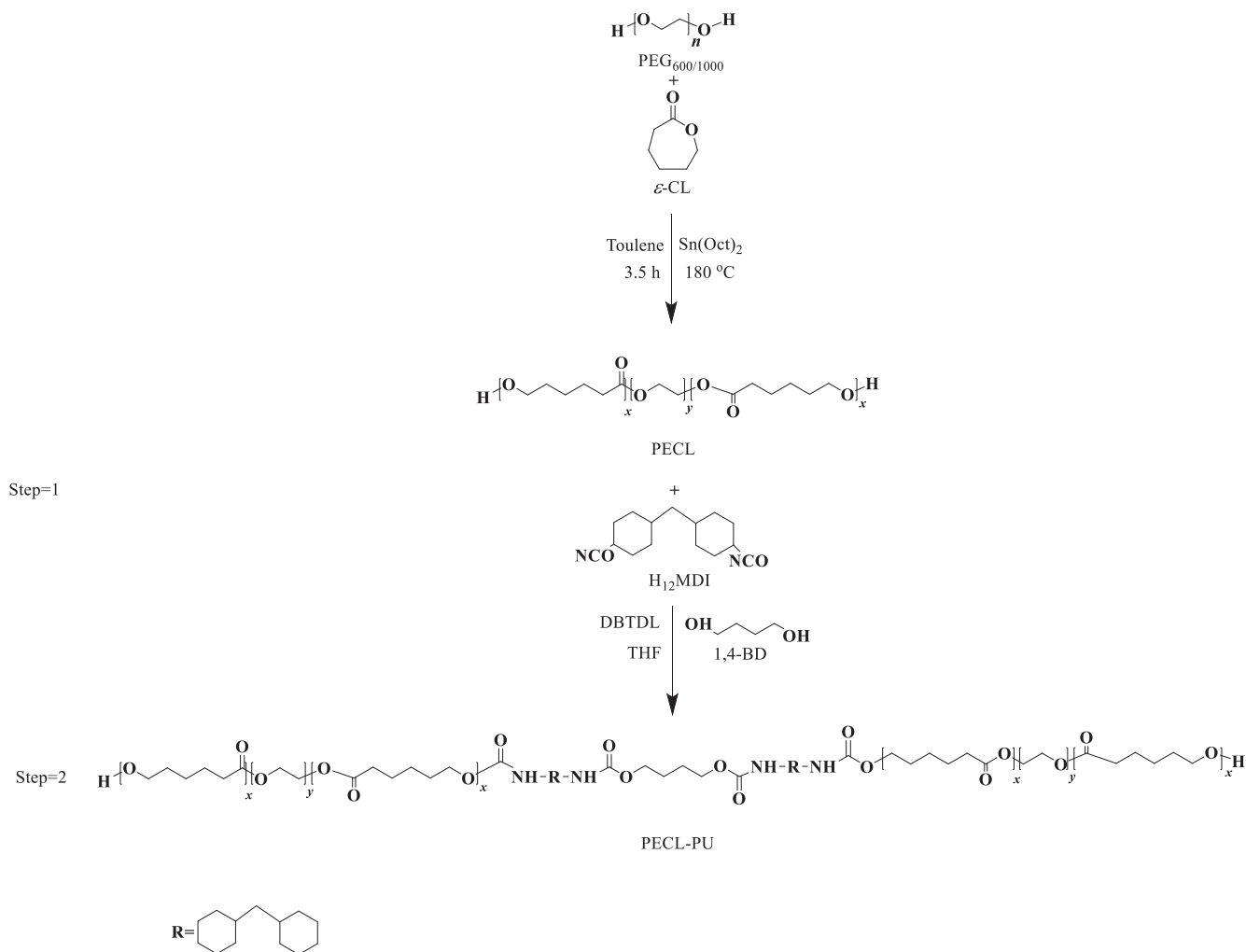


Fig. 1. Schematic illustration of the synthetic procedure of PECL-PU.

Table 1
Molar ratio and characterization data of PECL-PU

Sample	Feed molar ratio CL/PEG _{600/1000} /H ₁₂ MDI/1,4 BD	Polymer composition CL/PEG _{600/1000} /H ₁₂ MDI/1,4 BD ^①	$M_w^{②}$ / kDa	$M_n^{②}$ / kDa	$\bar{M}^{②}$	$T_m^{③}/^{\circ}\text{C}$	$\Delta H_m^{③}/$ $\text{J}\cdot\text{g}^{-1}$	Degree of crystallinity ^③ /%	$\gamma^{④}/\text{MPa}$	TGA ^⑤
PECL	46/3/3/0	37/63/0/0	19.8	7.4	2.6	44.0	40.5	29.8	–	286
CL-PU	46/0/14/14	71/0/19/10	99.0	46.1	2.1	59.7	45.0	33.1	2.43 ± 0.20	305
PECL-PU1	46/1/14/14	62/17/22/13	60.9	37.0	1.6	44.8	10.4	7.7	2.51 ± 0.40	342
PECL-PU2	46/2/14/14	54/22/15/9	78.7	48.6	1.6	38.6	7.2	5.3	1.39 ± 0.20	323
PECL-PU3	46/3/14/14	39/42/12/7	64.9	44.3	1.4	–	–	–	1.61 ± 0.50	320

^① Molar percentage determined by ¹H NMR.

^② Determined by GPC.

^③ Determined by DSC.

^④ Adhesion strength.

^⑤ Temperature at 50% mass loss as determined by TGA.

2.4. Preparation of PECL-PU coating

The glass slide was cut by the diamond cutter in a 25 mm × 25 mm slide; they were separately sonicated in ethanol in an ultrasonic bath for 15 min then washed twice with ethanol and deionized water. The coating solution was prepared by 1% polymer in THF. The solution was directly dispensed onto the glass slide on the spin coater. The glass slide was spun for the 60 s at

4000 r·min^{−1} with an accelerating period of 5 s. Afterward, the coating glass slide was kept at 40 °C temperature for 15 min to remove THF by evaporation to the surface, then carefully dried the glass surface. The coated glass slide was maintained at 40 °C for 15 min to allow THF to evaporate to the surface, and then the glass surface was thoroughly dried. The antibacterial, anti-diatom, and protein resistance tests were performed on the spin-coated coating.

3. Results and Discussion

The adhesion strength of carpolactone-co-polyurethane (CL-PU) and PECL-PU coating on the epoxy-based panel are depicted in Table 1, Fig. 2 and Fig. S4(a)–(c) (Supplementary Material). CL-PU had an adhesive strength of (2.43 ± 0.20) MPa. When PEG segments were introduced in the PECL-PU, the adhesion strength varied. PECL-PU1, PECL-PU2, and PECL-PU3 had adhesion strengths of (2.51 ± 0.40) , (1.39 ± 0.20) , and (1.61 ± 0.50) MPa, respectively. The results showed that PECL-PU1 had the greatest adhesive strength to the substrate. As a result, PECL-PU1 had a more flexible chain than CL-PU and other polymers, resulting in a high degree of polar contact and hydrogen bonding between epoxy-based substrates [45]. When the content of PEG segments were raised by more than 1 mmol, then polar interaction and hydrogen bonding between epoxy-based substrates became much weaker than CL-PU due to their hydrophilicity (Fig. 2). Additionally, Fig. S4(c) also showed that the failure problem of adhesion strength due to some reason. (1) The adhesion strength between adhesive agent and epoxy resin have collapsed. (2) Some area of epoxy resin on the substrate became collapse due to the pulling upper dolly opposite direction. Therefore we got failure result during the adhesion strength test.

The DSC thermograms of CL-PU, PECL-PU1, PECL-PU2, and PECL-PU3 are shown in Table 1 and Fig. 3. The endothermic peak (T_m) and melting enthalpy shift (ΔH_m) of the CL-PU were 59.7°C and $45.0\text{ J}\cdot\text{g}^{-1}$ due to the existence of crystals of various sizes of PCL in the polymer. PEG structure are typically semicrystalline and their incorporation into cross-linked network inhibit crystallization [43]. Therefore, the PEG segments interrupted the natural crystallization of PCL segments in the PECL-PU1, PECL-PU2, and PECL-PU3 chains. The T_m and ΔH_m of PECL-PU1 and PECL-PU2 were 44.8 , 38.6°C , and 10.4 , $7.2\text{ J}\cdot\text{g}^{-1}$ compared to CL-PU in the heating curve. For PECL-PU3, the melting peak was no longer reported in the heating curve, which revealed that the PECL-PU3 was ultimately more amorphous than CL-PU. The intensity of the characteristic peaks of PCL was significantly reduced as the PEG segments raised in the degradable polyurethane and the temperature of melting shifted towards the low point [46]. All the results indicated that the copolymerization of CL-PU with PEG segments reduced the crystallinity of soft segments. The decrease of crystallinity is because the PEG segment affected the irregularity of the spherulite size and crystallinity of the PCL in the polymer chain.

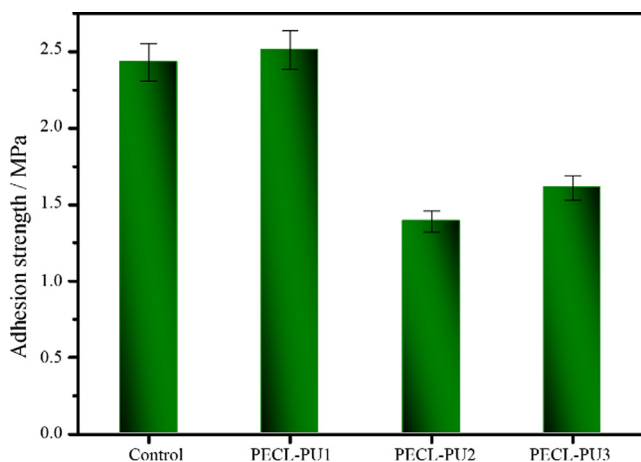


Fig. 2. The adhesion strength of PECL-PU with different PEG contents. Error bars were obtained from triplet measurement.

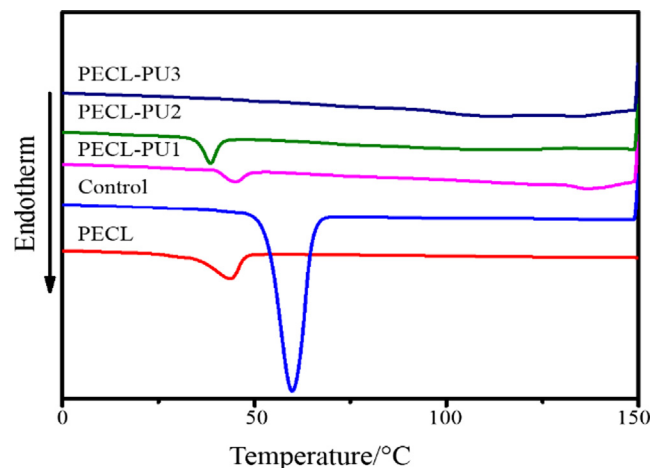


Fig. 3. DSC curve of CL-PU, PECL-PU1, PECL-PU2, and PECL-PU3.

The hydrolytic degradation experiment was carried out in ASW with the same pH and salt content as seawater, and the impact of microbe adsorption was ignored. The hydrolysis rate of the attached group and the diffusion of water kinetics within the polymer matrix must be balanced for the polymer to hydrolyze [47]. For 45 days, the hydrolytic degradation of films were observed, as shown in Fig. 4, which demonstrated the degradation efficiency of CL-PU and PECL-PU film in the ASW at 25°C . After three days, all PECL-PU chips began to mass loss. The CL-PU film's increased crystallinity and relative hydrophobicity exhibited a $(24.1 \pm 7.0)\text{ mg}\cdot\text{cm}^{-2}$ mass loss during the study, as confirmed by WCA, DSC, and XRD curve (Fig. S3). However, we assumed that the PCL's ester group mainly in the degradable polyurethane, was considered the initial cleavage in poly (ester urethane) during the degradation, the indication of ester degraded can be seen in Fourier-transform infrared spectroscopy (FTIR) analysis (Fig. S5). The polymer's crystallinity impeded water absorption since it dominated the entire hard portion practically. Hence, the water molecules barely entered the polymer matrix, degrading only the surface layer [48]. The percent of mass loss of PECL-PU1, PECL-PU2, and PECL-PU3 was attained at (35.0 ± 5.0) , (41.9 ± 7.0) , and $(52.1 \pm 6.0)\text{ mg}\cdot\text{cm}^{-2}$ after 45 days, respectively. PECL-PU1, PECL-PU2, and PECL-PU3 degraded more rapidly because PEG segments decreased PCL's crystallinity and spherulite size while also increasing the amorphous area in the hard segment, as confirmed by DSC (Fig. 3) and XRD curve (Fig. S3). The hydrolytic degradation of the PECL-PU film enhanced with increasing PEG segment concentration. PEG segments improved the irregularity of the PCL's spherulite size and resulted in increased hydrophilicity in ASW, as confirmed by immersion effect on wetting characteristic (Fig. S6(a)). The content of the PEG segments also controlled the degree of degradation. Biodegradable resins may be extremely antifouling if the biodegradation rate is high enough, but it comes with consequences: a high degradation rate does shorten the lifetime of the resin [49]. Antifouling coatings, which are present at the coating-water interface and may prevent fouling species from settling, are an excellent technique for increasing the antifouling ability of biodegradable polymers by integrating ecologically friendly compound [50].

Polymer biodegradation may be facilitated by enzymes secreted by microbial communities thriving in the marine environment. The hydrolytic degradation of PECL-PU was also studied in $0.5\text{ mg}\cdot\text{ml}^{-1}$ enzymatic media (lipase) and natural seawater at room temperature, as shown in Fig. 5. Enzymes are required for hydrolysis and then biodegradation. It has been shown that the CL-PU increases biodegradation through an enzyme released by numerous microor-

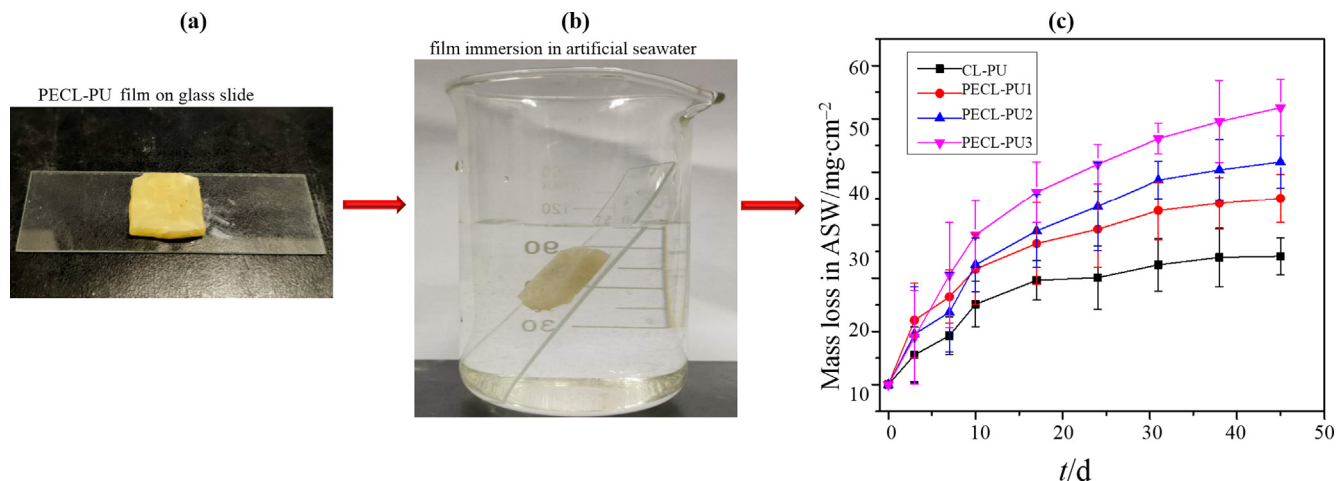


Fig. 4. Hydrolytic degradation performance. (a) The digital image (left side) for the fabrication of the PECL film (20 mm × 20 mm × 2 mm) on the glass slide, before incubation with ASW ($t = 0$). (b) Digital image showing the set up for mimicking static seawater (without enzyme) condition. The films were immersed vertically in ASW at room temperature for 45 days with the static condition. (c) The hydrolytic degradation of PECL-PU (mass loss of PECL-PU) at room temperature in 100 ml of ASW for 45 days ($t = 45$ d). Error bars were obtained from distinct measurements.

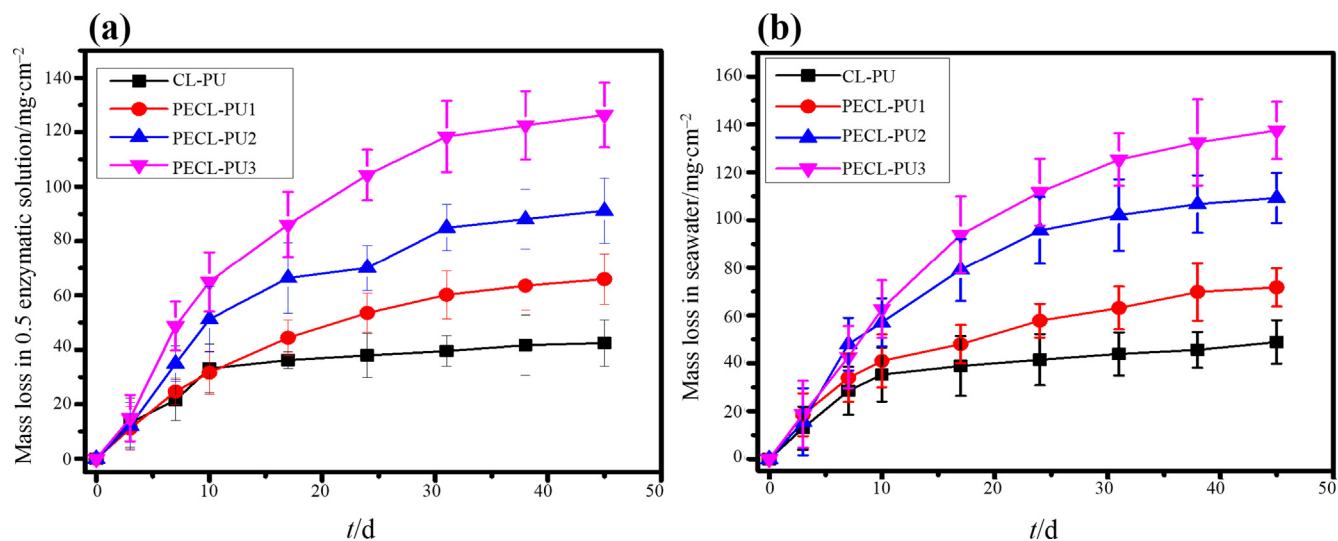


Fig. 5. The time dependence of enzymatic solution and seawater of CL-PU and PECL: (a) 0.5 mg·ml⁻¹ enzymatic solution and (b) seawater at room temperature. Error bars were obtained from triplet measurement.

ganisms in the sea. To catalyse hydrolysis, enzymes must interface with their substrate (*i.e.*, the ester bond of the polymer) [51]. CL-PU exhibited (42.5 ± 7.8) and (48.9 ± 9.1) mg·cm⁻² mass loss in enzymatic solution and seawater, respectively. The 0.5 mg·ml⁻¹ enzymatic solution and seawater both showed enhanced hydrolytic degradation of CL-PU. While PCL in the CL-PU backbone had high crystallinity and hydrophobicity, this has resulted in decreased interaction between esterase families and reduced enzymatic activity over time. In addition, the bio-degradation rate also effect on the polyester group of polymer due to their crystallinity, ester group density and spherulite size [49].

PECL-PU hydrolyzed considerably more rapidly than CL-PU. The PEG segments decreased the crystalline nature and enhanced the amorphous area in the PECL-PU polymer, which was facilitating enzyme attack. Water molecules entered deeply into the ester group of the biodegradable polyurethane structure through the PEG segments, leading to a significant hydrolysis rate. The mass loss of PECL-PU with different content of PEG segments was (65.9 ± 8.3), (91.2 ± 10.2), and (126.3 ± 13.5) mg·cm⁻² in

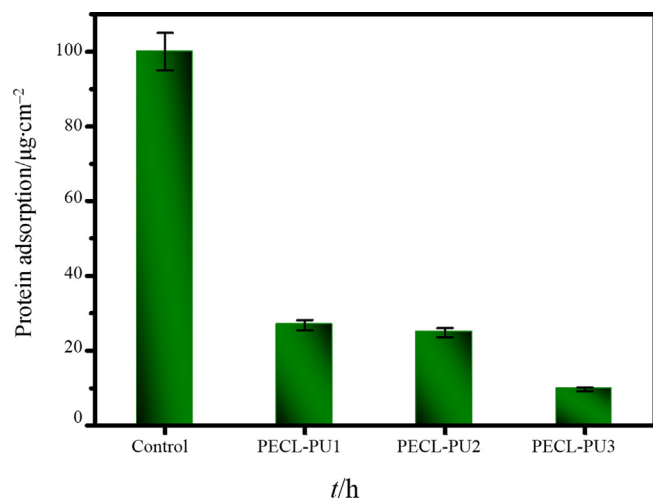
the 0.5 mg·ml⁻¹ enzymatic solution, and they also showed (71.7 ± 5.2), (109.3 ± 5.9) and (137.5 ± 8.2) mg·cm⁻² mass loss in the seawater. The biodegradation study indicated that the PECL-PU degraded more rapidly than the control. The hydrolytic degradation was quicker initially but subsequently became steady. Here, one small question raised that PEG is relatively known as generated repulsion of protein because of the adverse effect of PEG content on enzyme accessibility to the ester group, but it could be improved the hydrolytic degradation on a longer time.

We have also compared some biodegradable polymers-based literature to comprehensively evaluate the hydrolytic degradation performance of PECL-PU coating in Table 2. The copolymerization of ϵ -CL or PLA with other polymer or materials units was previously described as an approach for reducing crystallinity and increasing the degradability of biodegradable polyurethane. But our synthesized PECL-PU coating revealed promising hydrolytic degradation than PCL/TCPM-PU, PU-PCL-S336, PU-PLA-S336, PPG/CL-PU, PLA-PU-TSA, PEA-PU, PLLA-*b*-PDMS-PU, PCL-PU 80/rosin, and PLLA-FDEA-PU coatings. Moreover, others reported work

Table 2

A comparison of performance between biodegradable polymer/materials

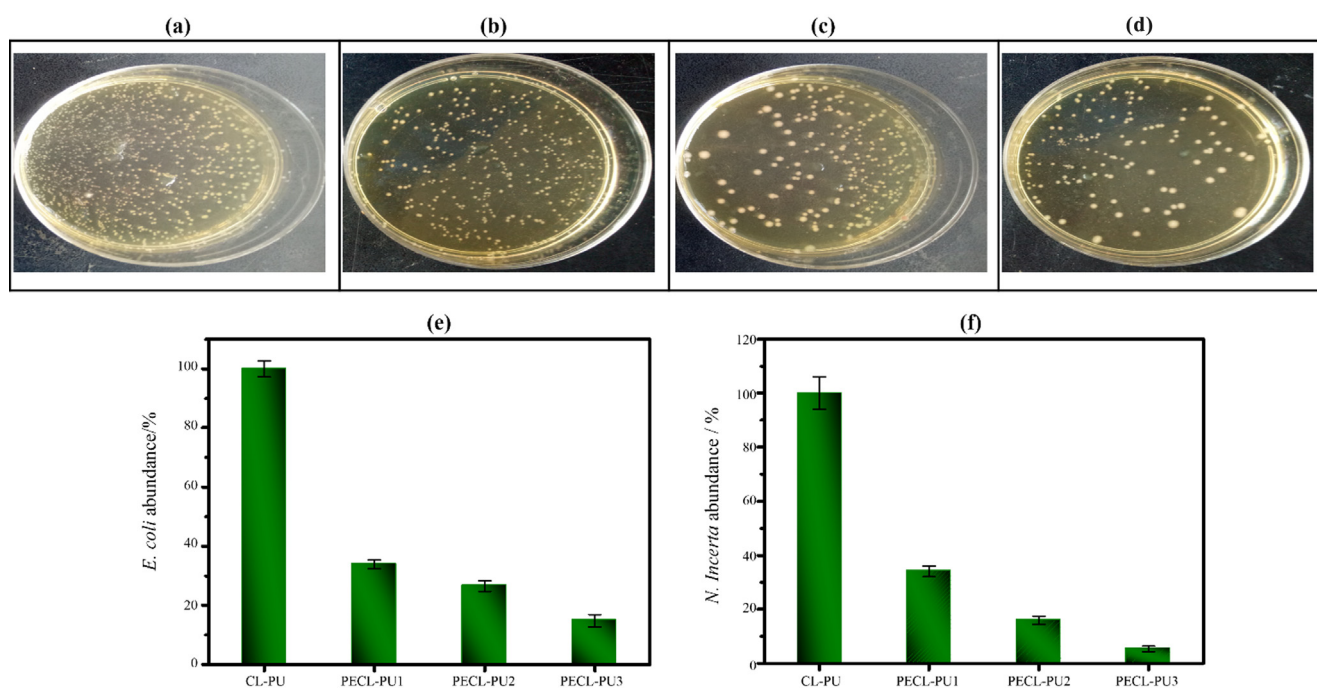
Biodegradable polymer/material	Hydrolytic degradation	Reference
PECL-PU	71.7–137.5 mg·cm ⁻² /45 d	This work
PCL/TCPM-PU	>0.05–0.15 mg·cm ⁻² /60 d	[48]
PU-PCL-S336	1.0 mg·ml ⁻² /90 d	[52]
PLA-PU-TSA	1.4–4.0 mg·cm ⁻² /60 d	[53]
PEA-PU	0.8 mg·cm ⁻² /60 d	[54]
PLLA-b-PDMS-PU	0.03 mg/60 d	[55]
PCL-PU 80/rosin	0.06 g/90 d	[56]
PLLA-FDEA-PU	0.0039–0.0140 g/100 d	[57]

**Fig. 6.** BSA protein (1 g·L⁻¹) adsorption on different coatings at 37 °C for 24 h. Error bars were obtained from triplet measurement.

'PEG/CL-PU' coating showed significantly higher hydrolytic degradation during 30 days; it means that the shelf life of this polymer is limited.

The antifouling property of the CL-PU and PECL-PU film was evaluated by protein resistance, an antibacterial, and anti-diatom assay. To evaluate the protein resistance, CL-PU and PECL-PU were collected on such target coated on the glass slide. After rinsing with PBS, these films were immersed into FTIC-labelled bovine serum albumin (BSA) as a model protein under 37 °C for 24 h. In addition, our coating CL-PU surface was used as a reference (Fig. 6). The surface of BSA concentration was (27.2 ± 2.1) μg·cm⁻². The surface of BSA concentration was reduced to (7.3 ± 1.6), (6.7 ± 1.2), and (2.6 ± 0.8) μg·cm⁻² after being modified with PEG segments, as shown in Fig. 6. The water CA analysis revealed a decrease in the contact angle from CL-PU to PECL-PU1, PECL-PU2, and PECL-PU3, which is confirmed by the WCA experiment. The amount of absorbed BSA on the PECL-PU3 films decreased by over 10 times lower than that on the control. The PECL-PU3 film had less CA than the control and other samples, indicating that the PEG-modified surface had been exposed to the aqueous solution, as confirmed by the immersion effect in WCA. PEG segments swell to cover the additional surface of the polymer; thus, it appears that the PECL-PU can inhibit protein adsorption as long as the PEG segments can completely cover the surface in its hydrated form.

To further investigate the antifouling properties of the degradable polyurethane through bacterial assay. The bacterial adhesion tests were conducted with *E. coli* cells as a model microorganism to adhere to the surface. The resistance of *E. coli* to adhesion on CL-PU and PECL-PU coated glass slides were investigated. The CL-PU was employed as control. Samples were incubated 1.0 × 10⁵ cells suspension of *E. coli* incubation for 14 h. The control had high coverage of *E. coli* cells, indicating that bacteria clustered together without PEG segments. The favorable interaction between the hydrophobic interface and bacteria's cell wall is always retained

**Fig. 7.** Digital images show the influence of the PEG segments on the growth of Gram-negative (*E.coli*) bacteria. After incubation for 24 h at 37 °C, the bacteria cell were plated in the appropriate medium (agar LB): (a) Control without PEG segments, (b) PECL-PU1, (c) PECL-PU2, and (d) PECL-PU3. (e) The abundance of *E. coli* cells after incubation in PBS buffer for 14 h with the PECL-PU3 and a CL-PU as control. (f) The absorbance of chlorophyll-a concentration of diatoms. Error bars were obtained from distinct measurements.

because it is made of peptidoglycan [58]. The number of bacteria adhering to the control was CL-PU was (973 ± 70) cells, which is defined as 100%, as shown in Fig. 7(a) and (e). PECL-PU1, PECL-PU2, and PECL-PU3 coated surfaces showed significantly lower bacteria attachment relative to the control. *E. coli* attachment was reduced by (330 ± 37) and (259 ± 28) cells, indicating a reduction of 66.1% for PECL-PU1 and 73.78% for PECL-PU2 coated surfaces (Fig. 7(b)–(d)). The incorporation of PEG had significantly repelled the attachment of bacteria to the surface and provided a hydrophilic defensive environment. Therefore they inhibited colonization of the bacteria on the surface.

Meanwhile, the PEG segments content in the polymer increased; thus, it decreased the number of bacteria settlement was observed (PECL-PU1 and PECL-PU2) owing to hydrophilicity. However, PECL-PU3 showed a significant decrease in the amount of bacteria cell absorbing with (144 ± 18) cells (Fig. 7(d)), which was 85.2% reduction in *E. coli* attachments compared to control and other samples, a significant increase in bacterial fouling resistance, as shown in Fig. 7(e). The extremely hydrated layer of PEG was inhibited bacteria from sticking to the polymer surface. Therefore, the PECL-PU3 surface exhibited a higher reduction of the bacteria colony than CL-PU. Because surface hydration is responsible for the improved antifouling capabilities, which are expected to

diminish nonspecific adsorption of protein released by bacteria. Notably, the PECL-PU coating maintains its effectiveness against *E. coli*, implying that this coating with biodegradation ability can be used in practical applications and systems in the future.

The adherence of *Navicula incerta* diatom cells to the CL-PU and PECL-PU surfaces was investigated. In contrast, CL-PU was employed as a positive control. The quantity of *Navicula incerta* diatoms adhering to the surface of control and PECL-PU1 after 7 days was revealed in Fig. 7(f) & 7S. Fig. 7(f) demonstrated the creation of (chlorophyll-a) on control with an absorbance of 0.779, indicating that control without PEG segments was unable to resist (chlorophyll-a) formation. PECL-PU1, PECL-PU2, and PECL-PU3 exhibited a high anti-diatom absorption capacity of 0.265, 0.124, and 0.042, respectively, indicating that PEG segments play a critical role in preventing the development of (chlorophyll-a) (aforementioned). Additionally, Fig. 7S demonstrated that the control sample had a greater number of diatoms attached than the other samples, indicating that CL-PU was insufficient resistance to diatom adhesion owing to its hydrophobicity (low hydrophilicity). However, the PECL-PU3 showed a reduced density of *Navicula incerta* diatoms (5.39%) on the surface due to their hydrophilicity, revealing that the PEG-modified surface effectively inhibited diatoms compared to the control. The PECL-PU1 and PECL-PU2 showed 34.01%

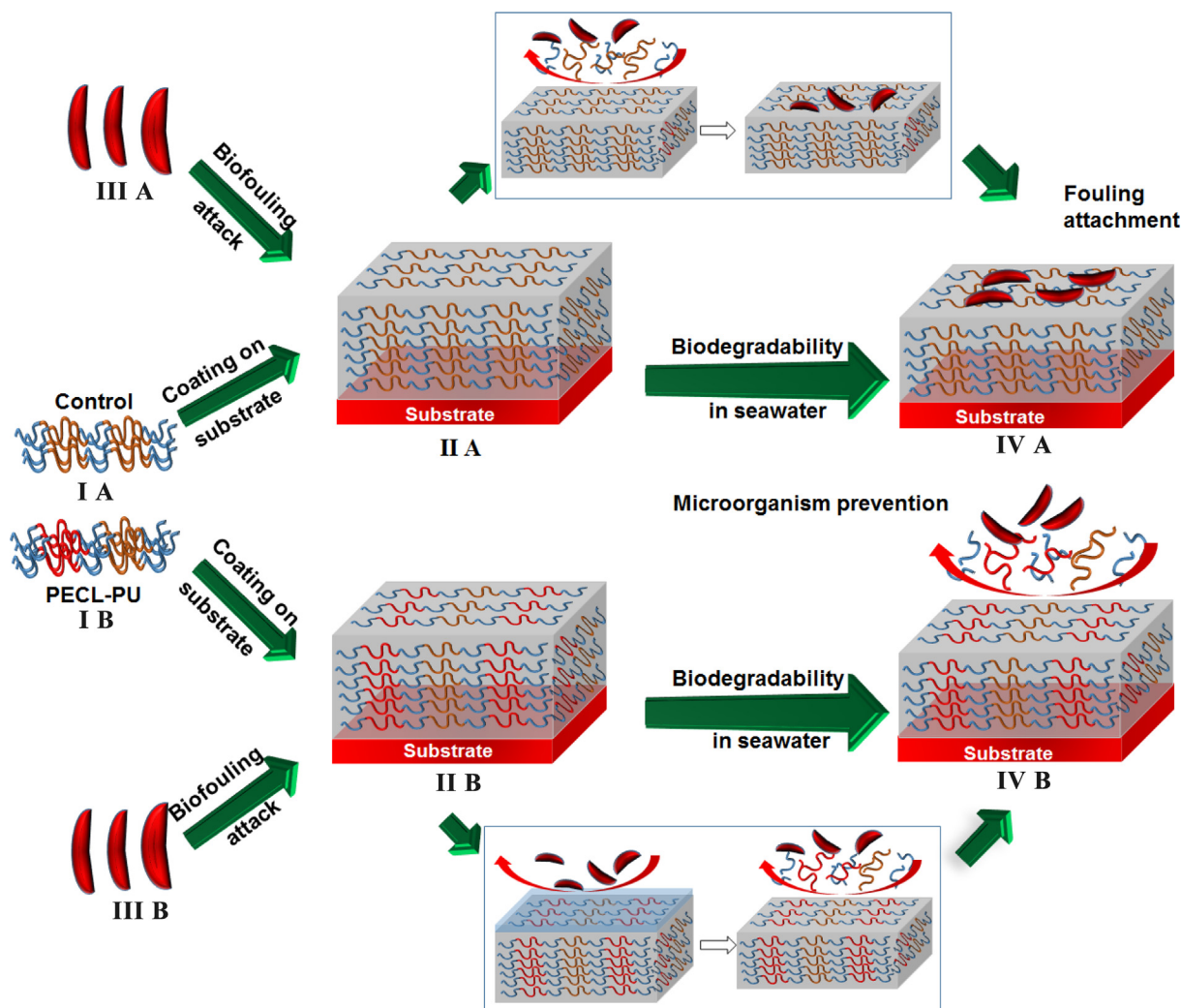


Fig. 8. Antifouling properties of biodegradable polyurethane. Schematic illustration of the (IA) CL-PU, (IB) PECL-PU, (IIA & IIB) the casting solution on the substrate, (IIIA & IIIB) the marine microorganism attack for the adhesion on the substrate, (IVA) the marine organism attaches on the surface due to very slow degradability in the seawater, and (IVB) the preventing of the microorganism during the biodegradation in the presence of PEG segments in the seawater for a long period.

and 15.91% diatom attached to the surface. The result indicated that the content of PEG segments improved the anti-diatom activity.

In seawater with pH $\sim$ 8.2 and higher salt content, even in the typical marine ecosystem, a massive number of microbes such as actinomycetes and bacteria may generate enzymes [59]. To measure enzymatic degradation in the marine environment, we observed polymer degradation in the presence of lipase and seawater. The rapid surface change improves maritime antifouling capability but a shorter service lifetime. Our study showed that the PECL-PU might degrade in the aquatic ecosystem, and its rate of degradation improves with the contents of PEG segments [58]. The PEG is chemically attached to a biodegradable polyurethane in this application. It is the breakage of the ester bond in the PCL segment and the urethane group (NHCOO) that is responsible for the hydrolysis reaction of PECL-PU coatings. The latter is typically stable, although its cleavage may occur under certain conditions such as alkali or fungal [24]. The addition of the PEG segments can boost the reducing microbial activity (aforementioned) and also regulate the degradation of PECL-PU. The reducing bacterial and diatom activity of PECL-PU is consistent with its protein resistance ability. Because bacterial adhesion is often initiated by the adsorption of extracellular polymeric substances (EPS), which are predominantly made of glycoproteins. In the marine environment, a protein-resistant surface may reduce the initial adhesive strength and inhibit microbial colonization [26,60,61]. PEG segments based on degradable coating can be used as environment-friendly in marine antifouling activity. PECL-PU may degrade into an oligomer and subsequently into tiny fragments that eventually decompose into water and carbon dioxide [62]. Therefore the coating is no negative impact for the sea-environment.

4. Proposed Mechanism

Here, we can propose a potential mechanism according to the above results. This CL-PU coating is linked with PEG segments, which can prevent adhesion of biofouling and higher reduction of crystallinity in the polymer. In the marine environment, the hydrolytic degradation of biodegradable polyurethane-based coating can include the cleavage of the ester bond in the PCL and urethane segments (NHCOO). The fungus secretes enzymes (lipase, protease, etc.) in the marine environment that may cause cleavage of the CL-PU and PECL-PU. An enzyme attacking the amorphous region of the CL-PU creates additional crystals, so biofouling attaches to the surface of CL-PU due to sluggish hydrolytic degradation with high crystallinity in the marine environment (Fig. 8A). The PEG segments chemically attach to the biodegradable polyurethane. Thus, the content of PEG segments can decrease the crystallinity of the PCL in the polymer and make a hydration layer on the surface, preventing the biofouling attachment on the substrate in the marine environment. The content of PEG segments also regulates the degradation rate of PECL-PU with antifouling activity (Fig. 8B). As a result of the constant occurrence of hydrolytic and enzymatic degradation, we assume that the proposed mechanism would be the ultimate goal of the remarkable efficiency of PECL-PU coating even in the complex marine environment.

5. Conclusions

In this work, we successfully prepared a biodegradable polyurethane using PCL, PEG segments, and H₁₂MDI and 1, 4 BD, whose preparation was facile, up scalable, and inexpensive. Our work shows that the introduction of PEG segments reduced the crystallinity and increased the amorphous region in the polymer. The existence of ester units into the main chain of the biodegradable

polymer and the degradation rate of such degradable polymer was sensitive to enzyme concentration, which contributed to the self-renewing dynamic surfaces. Moreover, the PEG based hydration layer induced by hydrolysis exhibited promising protein resistance, antibacterial activity, and anti-diatom performance. The experimental results demonstrated that the synthesized degradable polyurethane with PEG segments could inhibit the formation of biological condition films without being harmful to marine organisms.

Data Availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (21878267).

Supplementary Material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.cjche.2022.09.004>.

References

- [1] H. Yan, Q.S. Wu, C.M. Yu, T.Y. Zhao, M.J. Liu, Recent progress of biomimetic antifouling surfaces in marine, *Adv. Mater. Interf.* 7 (20) (2020) 2000966.
- [2] X.Y. Hu, J.H. Tian, C. Li, H. Su, R.R. Qin, Y.F. Wang, X. Cao, P. Yang, Amyloid-like protein aggregates: A new class of bioinspired materials merging an interfacial anchor with antifouling, *Adv. Mater.* 32 (23) (2020) 2000128.
- [3] A. Vena, S. Kolle, S. Stafslien, J. Aizenberg, P. Kim, Self-stratifying porous silicones with enhanced liquid infusion and protective skin layer for biofouling prevention, *Adv. Mater. Interf.* 8 (2) (2021) 2000359.
- [4] M.S. Selim, S.A. El-Safty, M.A. Shenashen, S.A. Higazy, A. Elmarakbi, Progress in biomimetic leverages for marine antifouling using nanocomposite coatings, *J. Mater. Chem. B* 8 (17) (2020) 3701–3732.
- [5] S. Kinkley, J. Helmuth, J.K. Polansky, I. Dunkel, G. Gasparoni, S. Fröhler, W. Chen, J. Walter, A. Hamann, H.R. Chung, reChIP-seq reveals widespread bivalency of H3K4me3 and H3K27me3 in CD4⁺ memory T cells, *Nat. Commun.* 7 (2016) 12514.
- [6] M.S. Selim, M.A. Shenashen, S.A. El-Safty, S.A. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog. Mater. Sci.* 87 (2017) 1–32.
- [7] M. Lejars, A. Margaillan, C. Bressy, Fouling release coatings: A nontoxic alternative to biocidal antifouling coatings, *Chem. Rev.* 112 (8) (2012) 4347–4390.
- [8] R. Ciriminna, V. Pandarus, G. Gingras, F. Béland, M. Pagliaro, Closing the organosilicon synthetic cycle: Efficient heterogeneous hydrosilylation of alkenes over SiliaCatPt(0), *ACS Sustain. Chem. Eng.* 1 (2) (2013) 249–253.
- [9] X. Su, D.Z. Hao, Z.N. Li, X.L. Guo, L. Jiang, Design of hierarchical comb hydrophilic polymer brush (HCHPB) surfaces inspired by fish mucus for anti-biofouling, *J. Mater. Chem. B* 7 (8) (2019) 1322–1332.
- [10] H. Okoro, O. Fatoki, F.A. Adekola, B.J. Ximba, R. Snyman, Sources, environmental levels and toxicity of organotin in marine environment: A review, *Asian J. Chem.* 23 (2) (2011) 473–482.
- [11] K.V. Thomas, S. Brooks, The environmental fate and effects of antifouling paint biocides, *Biofouling* 26 (1) (2010) 73–88.
- [12] R. Deng, T. Shen, H.L. Chen, J.X. Lu, H.C. Yang, W.H. Li, Slippery liquid-infused porous surfaces (SLIPSs): A perfect solution to both marine fouling and corrosion?, *J. Mater. Chem. A* 8 (16) (2020) 7536–7547.
- [13] C.Q. Wei, G.F. Zhang, Q.H. Zhang, X.L. Zhan, F.Q. Chen, Silicone oil-infused slippery surfaces based on sol-gel process-induced nanocomposite coatings: A facile approach to highly stable bioinspired surface for biofouling resistance, *ACS Appl. Mater. Interf.* 8 (50) (2016) 34810–34819.
- [14] Q.Y. Xie, H.H. Zeng, Q.M. Peng, C. Bressy, C.F. Ma, G.Z. Zhang, Self-stratifying silicone coating with nonleaching antifoulant for marine anti-biofouling, *Adv. Mater. Interf.* 6 (13) (2019) 1900535.
- [15] P. Hu, Q.Y. Xie, C.F. Ma, G.Z. Zhang, Silicone-based fouling-release coatings for marine antifouling, *Langmuir* 36 (9) (2020) 2170–2183.

- [16] X.L. Zhan, G.F. Zhang, Q.H. Zhang, F.Q. Chen, Preparation, surface wetting properties, and protein adsorption resistance of well-defined amphiphilic fluorinated diblock copolymers, *J. Appl. Polym. Sci.* 131 (23) (2014) 41167.
- [17] F. Gao, G.F. Zhang, Q.H. Zhang, X.L. Zhan, F.Q. Chen, Improved antifouling properties of poly(ether sulfone) membrane by incorporating the amphiphilic comb copolymer with mixed poly(ethylene glycol) and poly(dimethylsiloxane) brushes, *Ind. Eng. Chem. Res.* 54 (35) (2015) 8789–8800.
- [18] X. Chen, G.F. Zhang, Q.H. Zhang, X.L. Zhan, F.Q. Chen, Preparation and performance of amphiphilic polyurethane copolymers with capsaicin-mimic and PEG moieties for protein resistance and antibacteria, *Ind. Eng. Chem. Res.* 54 (15) (2015) 3813–3820.
- [19] G.X. Dai, Q.Y. Xie, X.Q. Ai, C.F. Ma, G.Z. Zhang, Self-generating and self-renewing zwitterionic polymer surfaces for marine anti-biofouling, *ACS Appl. Mater. Interf.* 11 (44) (2019) 41750–41757.
- [20] A. Ali, Y. Xiao, L.N. Song, J.K. Hu, Q.Q. Rao, M. Shoaib, B.U. Amin, X.L. Zhan, Q.H. Zhang, Biodegradable polyurethane based clay composite and their anti-biofouling properties, *Colloids Surf. A Physicochem. Eng. Aspects* 625 (2021) 126946.
- [21] Y.K. Demirel, M. Khorasanchi, O. Turan, A. Incek, On the importance of antifouling coatings regarding ship resistance and powering, In: 3rd International Conference on Technologies, Operations, Logistics and Modelling for Low Carbon Shipping, UK, 2013.
- [22] I.M. Organization, I.M.O.M.E.P. Committee, Anti-fouling systems: International Convention on the Control of Harmful Anti-fouling Systems on Ships, 2001 (AFS 2001) and Guidelines for Survey and Certification of Anti-fouling Systems on Ships (resolution MEPC. 102 (48)), Guidelines for Brief Sampling of Anti-fouling Systems on Ships (resolution MEPC. 104 (49)), and Guidelines for Inspection of Anti-fouling Systems on Ships (resolution, in: MEPC. 105, IMO Publishing, 2005.
- [23] A. Ali, M.I. Jamil, J.X. Jiang, M. Shoaib, B.U. Amin, S.Z. Luo, X.L. Zhan, F.Q. Chen, Q.H. Zhang, An overview of controlled-biode-release coating based on polymer resin for marine antifouling applications, *J. Polym. Res.* 27 (4) (2020) 1–17.
- [24] Q.Y. Xie, J.S. Pan, C.F. Ma, G.Z. Zhang, Dynamic surface antifouling: Mechanism and systems, *Soft Matter* 15 (6) (2019) 1087–1107.
- [25] S.S. Chen, C.F. Ma, G.Z. Zhang, Biodegradable polymers for marine antifouling: Poly(ϵ -caprolactone)/poly(butylene succinate) blend as controlled release system of organic antifoulant, *Polymer* 90 (2016) 215–221.
- [26] J.L. Ma, C.F. Ma, G.Z. Zhang, Degradable polymer with protein resistance in a marine environment, *Langmuir* 31 (23) (2015) 6471–6478.
- [27] Q.Y. Xie, Q.N. Xie, J.S. Pan, C.F. Ma, G.Z. Zhang, Biodegradable polymer with hydrolysis-induced zwitterions for antibiofouling, *ACS Appl. Mater. Interf.* 10 (13) (2018) 11213–11220.
- [28] Y.F. Wang, Q.F. Hong, Y.J. Chen, X.X. Lian, Y.F. Xiong, Surface properties of polyurethanes modified by bioactive polysaccharide-based polyelectrolyte multilayers, *Colloids Surf. B Biointerf.* 100 (2012) 77–83.
- [29] R. Konradi, B. Pidhatika, A. Mühlebach, M. Textor, Poly-2-methyl-2-oxazoline: A peptide-like polymer for protein-repellent surfaces, *Langmuir* 24 (3) (2008) 613–616.
- [30] G. Emilsson, R.L. Schoch, L. Feuz, F. Höök, R.Y.H. Lim, A.B. Dahlin, Strongly stretched protein resistant poly(ethylene glycol) brushes prepared by grafting-to, *ACS Appl. Mater. Interf.* 7 (14) (2015) 7505–7515.
- [31] S.Y. Jiang, Z.Q. Cao, Ultralow-fouling, functionalizable, and hydrolyzable zwitterionic materials and their derivatives for biological applications, *Adv. Mater.* 22 (9) (2010) 920–932.
- [32] J. Wu, C. Zhao, R.D. Hu, W.F. Lin, Q.M. Wang, J. Zhao, S.M. Bilinovich, T.C. Leeper, L.Y. Li, H.M. Cheung, S.F. Chen, J. Zheng, Probing the weak interaction of proteins with neutral and zwitterionic antifouling polymers, *Acta Biomater.* 10 (2) (2014) 751–760.
- [33] W. Yandi, S. Mieszkina, A. di Fino, P. Martin-Tanchereau, M.E. Callow, J.A. Callow, L. Tyson, A.S. Clare, T. Ederth, Charged hydrophilic polymer brushes and their relevance for understanding marine biofouling, *Biofouling* 32 (6) (2016) 609–625.
- [34] J. Koc, T. Simovich, E. Schönmann, A. Chilkoti, H. Gardner, G.W. Swain, K. Hunsucker, A. Laschewsky, A. Rosenhahn, Sediment challenge to promising ultra-low fouling hydrophilic surfaces in the marine environment, *Biofouling* 35 (4) (2019) 454–462.
- [35] C.F. Ma, H.J. Yang, X. Zhou, B. Wu, G.Z. Zhang, Polymeric material for anti-biofouling, *Colloids Surf. B Biointerf.* 100 (2012) 31–35.
- [36] C.H. Tsou, H.T. Lee, H.A. Tsai, H.J. Cheng, M.C. Suen, Synthesis and properties of biodegradable polycaprolactone/polyurethanes by using 2,6-pyridinedimethanol as a chain extender, *Polym. Degrad. Stab.* 98 (2) (2013) 643–650.
- [37] C.H. Tsou, H.T. Lee, W.S. Hung, C.C. Wang, C.C. Shu, M.C. Suen, M. de Guzman, Synthesis and properties of antibacterial polyurethane with novel bis(3-pyridinemethanol) silver chain extender, *Polymer* 85 (2016) 96–105.
- [38] G.R. da Silva, A. da Silva-Cunha Jr, F. Behar-Cohen, E. Ayres, R.L. Oréfice, Biodegradation of polyurethanes and nanocomposites to non-cytotoxic degradation products, *Polym. Degrad. Stab.* 95 (4) (2010) 491–499.
- [39] M.H. Xiao, N. Zhang, J. Zhuang, Y.C. Sun, F. Ren, W.W. Zhang, Z.S. Hou, Degradable poly(ether-ester-urethane)s based on well-defined aliphatic diurethane diisocyanate with excellent shape recovery properties at body temperature for biomedical application, *Polymers* 11 (6) (2019) 1002.
- [40] S.Z. Fu, L.L. Yang, J. Fan, Q.L. Wen, S. Lin, B.Q. Wang, L.L. Chen, X.H. Meng, Y. Chen, J.B. Wu, *In vitro* mineralization of hydroxyapatite on electrospun poly(ϵ -caprolactone)-poly(ethylene glycol)-poly(ϵ -caprolactone) fibrous scaffolds for tissue engineering application, *Colloids Surf. B Biointerfaces* 107 (2013) 167–173.
- [41] Y. Li, H.M. Chen, D. Liu, W.X. Wang, Y. Liu, S.B. Zhou, pH-responsive shape memory poly(ethylene glycol)-poly(ϵ -caprolactone)-based polyurethane/cellulose nanocrystals nanocomposite, *ACS Appl. Mater. Interf.* 7 (23) (2015) 12988–12999.
- [42] H.P. Wang, D.J. Tong, L. Wang, L. Chen, N. Yu, Z.X. Li, A facile strategy for fabricating PCL/PEG block copolymer with excellent enzymatic degradation, *Polym. Degrad. Stab.* 140 (2017) 64–73.
- [43] L.D. Zhu, K.M. Bratlie, Supramolecular assemblies of alkane functionalized polyethylene glycol copolymers for drug delivery, *Mater. Sci. Eng. C* 81 (2017) 432–442.
- [44] Q.N. Xie, X. Zhou, C.F. Ma, G.Z. Zhang, Self-cross-linking degradable polymers for antifouling coatings, *Ind. Eng. Chem. Res.* 56 (18) (2017) 5318–5324.
- [45] American Society for Testing and Materials, Standard Practice for the Preparation of Substitute Ocean Water, ASTM International, 2013.
- [46] A. Güney, C. Gardiner, A. McCormack, J. Malda, D.W. Grijpma, Thermoplastic PCL-*b*-PEG-*b*-PCL and HDI polyurethanes for extrusion-based 3D-printing of tough hydrogels, *Bioengineering* 5 (4) (2018) 99.
- [47] A. Ali, L.N. Song, J.K. Hu, J.X. Jiang, Q.Q. Rao, M. Shoaib, S. Fahad, Y.J. Cai, X.L. Zhan, F.Q. Chen, Q.H. Zhang, Synthesis and characterization of caprolactone based polyurethane with degradable and antifouling performance, *Chin. J. Chem. Eng.* 34 (2021) 299–306.
- [48] J.L. Ma, C.F. Ma, Y. Yang, W.T. Xu, G.Z. Zhang, Biodegradable polyurethane carrying antifoulants for inhibition of marine biofouling, *Ind. Eng. Chem. Res.* 53 (32) (2014) 12753–12759.
- [49] C.F. Ma, L.G. Xu, W.T. Xu, G.Z. Zhang, Degradable polyurethane for marine anti-biofouling, *J. Mater. Chem. B* 1 (24) (2013) 3099–3106.
- [50] F. Faÿ, I. Linossier, J.J. Peron, V. Langlois, K. Vallée-Rehel, Antifouling activity of marine paints: Study of erosion, *Prog. Org. Coat.* 60 (3) (2007) 194–206.
- [51] E. Guégain, J.P. Michel, T. Boissenot, J. Nicolas, Tunable degradation of copolymers prepared by nitroxide-mediated radical ring-opening polymerization and point-by-point comparison with traditional polyesters, *Macromolecules* 51 (3) (2018) 724–736.
- [52] C.F. Ma, W.T. Xu, J.S. Pan, Q.Y. Xie, G.Z. Zhang, Degradable polymers for marine antibiofouling: Optimizing structure to improve performance, *Ind. Eng. Chem. Res.* 55 (44) (2016) 11495–11501.
- [53] J. Pan, Q. Xie, H. Chiang, Q. Peng, P.Y. Qian, C. Ma, G. Zhang, “From the nature for the nature”: An eco-friendly antifouling coating consisting of poly(lactic acid)-based polyurethane and natural antifoulant, *ACS Sustain. Chem. Eng.* 8 (3) (2019) 1671–1678.
- [54] S.S. Chen, C.F. Ma, G.Z. Zhang, Biodegradable polymer as controlled release system of organic antifoulant to prevent marine biofouling, *Prog. Org. Coat.* 104 (2017) 58–63.
- [55] B.L. Ou, M.L. Chen, Y.J. Guo, Y.H. Kang, Y. Guo, S.G. Zhang, J.H. Yan, Q.Q. Liu, D.X. Li, Preparation of novel marine antifouling polyurethane coating materials, *Polym. Bull.* 75 (11) (2018) 5143–5162.
- [56] C.F. Ma, W.P. Zhang, G.Z. Zhang, P.Y. Qian, Environmentally friendly antifouling coatings based on biodegradable polymer and natural antifoulant, *ACS Sustain. Chem. Eng.* 5 (7) (2017) 6304–6309.
- [57] M.L. Chen, B.L. Ou, Y.J. Guo, Y. Guo, Y.H. Kang, H.Y. Liu, J.H. Yan, L. Tian, Preparation of an environmentally friendly antifouling degradable polyurethane coating material based on medium-length fluorinated diols, *J. Macromol. Sci. A* 55 (6) (2018) 483–488.
- [58] T. Ekblad, G. Bergström, T. Ederth, S.L. Conlan, R. Mutton, A.S. Clare, S. Wang, Y. L. Liu, Q. Zhao, F. D'Souza, G.T. Donnelly, P.R. Willemsen, M.E. Pettitt, M.E. Callow, J.A. Callow, B. Liedberg, Poly(ethylene glycol)-containing hydrogel surfaces for antifouling applications in marine and freshwater environments, *Biomacromolecules* 9 (10) (2008) 2775–2783.
- [59] M. Ulbricht, H. Yang, Porous polypropylene membranes with different carboxyl polymer brush layers for reversible protein binding via surface-initiated graft copolymerization, *Chem. Mater.* 17 (10) (2005) 2622–2631.
- [60] G. Cheng, H. Xue, Z. Zhang, S.F. Chen, S.Y. Jiang, A switchable biocompatible polymer surface with self-sterilizing and nonfouling capabilities, *Angew. Chem. Int. Ed.* 47 (46) (2008) 8831–8834.
- [61] T. McPherson, A. Kidane, I. Szleifer, K. Park, Prevention of protein adsorption by tethered poly(ethylene oxide) layers: Experiments and single-chain mean-field analysis, *Langmuir* 14 (1) (1998) 176–186.
- [62] J.H. Yao, Z.W. Dai, J. Yi, H.L. Yu, B. Wu, L.Y. Dai, Degradable polyurethane based on triblock polyols composed of polypropyleneglycol and ϵ -caprolactone for marine antifouling applications, *J. Coat. Technol. Res.* 17 (4) (2020) 865–874.